

GLACIER PARK, MT, USA

WMO: 727790

Lat: 48.304N

Lon: 114.264W

Elev: 2957

StdP: 13.19

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24146

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.5	1.9	-13.1	3.0	-2.5	-6.5	4.3	3.4	25.3	18.2	20.8	18.8	3.7	10	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	90.4	62.4	87.2	61.5	83.8	60.5	65.1	83.2	63.3	81.6	61.8	80.0	7.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.1	83.6	69.4	57.0	77.2	67.7	55.0	71.9	66.5	31.4	83.5	30.0	81.3	28.9	80.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	17.5	15.3	-14.3	95.5	7.3	3.0	-19.6	97.7	-23.9	99.4	-28.0	101.1	-33.3	103.3
			-14.7	69.0	7.2	2.5	-19.9	70.8	-24.2	72.3	-28.2	73.8	-33.5	75.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.8	24.4	26.7	34.9	43.2	52.0	58.1	65.9	64.5	55.1	42.3	32.5	24.4	
	DBStd	16.55	10.50	9.75	7.96	6.32	6.59	6.28	5.65	5.58	6.73	7.01	8.30	9.74	
	HDD50	3797	795	653	467	219	56	8	0	0	26	252	527	794	
	HDD65	7921	1260	1073	932	653	405	222	58	78	302	703	976	1259	
	CDD50	1520	0	0	0	16	116	251	493	449	180	14	1	0	
	CDD65	168	0	0	0	0	1	14	85	62	6	0	0	0	
	CDH74	4363	0	0	0	7	116	422	1930	1564	322	2	0	0	
	CDH80	1620	0	0	0	0	17	117	792	613	81	0	0	0	
Wind	WSAvg	5.0	3.9	4.7	5.7	6.7	6.4	5.6	5.2	5.1	4.7	4.3	4.2	3.7	
Precipitation	PrecAvg	17.2	1.5	1.1	1.1	1.2	2.0	2.5	1.2	1.3	1.3	1.0	1.4	1.7	
	PrecMax	25.2	3.4	2.0	3.0	2.6	4.8	5.7	6.0	4.2	4.0	2.5	3.3	4.7	
	PrecMin	10.5	0.2	0.0	0.1	0.3	0.2	0.5	0.0	0.0	0.0	0.0	0.3	0.5	
	PrecStd	3.6	0.7	0.5	0.5	0.6	1.1	1.3	1.1	1.1	0.9	0.6	0.8	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	49.7	61.6	74.1	82.5	89.0	94.9	93.2	87.3	71.1	56.4	46.9	
		MCWB	39.0	41.7	46.3	52.7	58.1	63.0	64.0	62.4	60.2	54.3	48.8	41.5	
	2%	DB	41.0	43.7	54.3	67.8	77.2	83.0	91.5	89.8	81.9	64.9	51.2	41.8	
		MCWB	36.8	37.9	42.8	50.4	56.4	60.5	63.1	61.6	58.5	51.3	44.3	38.0	
	5%	DB	38.7	40.7	50.4	62.1	72.9	78.9	88.2	86.8	77.6	59.6	47.6	38.8	
		MCWB	35.2	35.9	41.2	47.3	54.5	59.0	62.3	60.8	57.0	48.4	42.3	35.4	
	10%	DB	36.8	38.3	46.9	56.8	68.1	74.6	84.6	83.3	72.7	55.1	44.4	36.3	
		MCWB	34.0	34.3	39.2	44.7	52.3	57.2	61.4	59.9	55.4	46.4	40.0	33.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.9	42.2	47.4	54.1	60.3	65.3	68.6	66.1	62.5	56.4	50.8	42.0
			MCDB	43.1	47.1	57.6	69.9	76.6	84.0	86.1	83.7	78.7	68.2	55.1	45.9
2%		WB	37.4	38.9	44.5	51.1	57.8	62.4	66.1	63.7	60.1	52.6	45.6	38.4	
		MCDB	40.1	42.7	52.6	65.1	73.6	79.3	84.1	82.9	77.6	61.9	50.2	41.2	
5%		WB	35.8	36.5	42.1	48.4	55.7	60.4	64.2	62.2	58.2	50.1	43.0	36.0	
		MCDB	38.2	39.6	48.8	60.0	69.9	75.4	82.4	81.6	74.9	57.5	47.0	38.2	
10%		WB	34.3	34.8	39.9	45.9	53.5	58.5	62.6	60.9	56.1	47.6	40.6	34.1	
		MCDB	36.3	37.7	45.8	55.9	65.4	71.7	80.6	79.8	70.9	54.2	44.0	35.9	
Mean Daily Temperature Range	5% DB	MDBR	12.9	15.6	18.6	22.3	24.6	25.2	32.8	33.5	29.4	22.7	14.6	11.7	
		MCDBR	13.4	17.1	25.5	33.5	35.1	35.3	39.7	39.9	39.0	31.2	19.4	12.6	
		MCWBR	10.2	12.1	15.7	17.8	16.5	15.6	15.3	15.7	17.9	18.5	13.7	10.0	
	5% WB	MCDBR	12.2	15.6	22.7	30.2	31.4	31.4	33.9	34.6	35.9	26.8	17.4	11.9	
		MCWBR	9.6	11.5	14.9	16.7	15.4	14.5	14.4	14.3	17.1	16.3	12.7	9.8	
Clear-Sky Solar Irradiance	taub	0.251	0.260	0.280	0.319	0.325	0.329	0.333	0.344	0.318	0.293	0.275	0.251		
	taud	2.395	2.374	2.416	2.408	2.493	2.523	2.518	2.482	2.530	2.594	2.530	2.420		
	Ebn at Noon	267	290	300	296	297	295	292	285	284	276	257	252		
	Edh at Noon	21	27	30	34	33	32	32	31	27	22	18	18		
All-Sky Solar Radiation	RadAvg	345	639	994	1421	1730	1889	2243	1848	1294	764	379	269		
	RadStd	39	65	106	122	138	178	164	127	134	84	34	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page